

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE Final Examination

Type: Theory

Course Code: CSE-2423 Database Management System

Total marks: 60

[Figures in the right-hand margin indicate full marks]

Course Outcomes and Bloom's taxonomy levels are mentioned in additional columns]

[SEGMENT-4]

Integrity, Security and Relational Database Design:

Domain constraint, Integrity, Assertions, Triggers, Authorization, Authentication, Security, Privileges, Roles, and Audit trails, Encryption-Decryption Algorithm, Decomposition etc.

Group-A			
		Sem	Marks
1.			
a)	Consider the following relational database: Student (sid, sname, email, dept_id, dob) Course (cid, cname, credits) Enrollment (sid, cid, enroll_date) Instructor (iid, iname, designation, salary) Teaches (iid, cid, semester) Write the SQL DDL for this schema, incorporating the following constraints: • sname, cname, and iname must NOT NULL • email must be unique and properly formatted • credits and salary must be greater than 0 • dept_id must exist in a referenced Department table (you may assume its schema) OR Why is normalization very important in DBMS? Consider the following set FD of functional dependencies for relation schema X = (A, B, C, D) FD: { $A \rightarrow B$, $B \rightarrow C$ } Transform schema X into 2NF, showing that the decomposition is a functional dependency preserving and lossless join decomposition.	Sp-25	6

b)	Given the following tables: Instructor (iid, iname, designation, salary) Teaches (iid, cid, semester) Write an assertion for the above schema to ensure that an instructor cannot teach more than two courses in the same semester . OR What is the difference between authorization and authentication ? Discuss how assertions differ from triggers .	Sp-25	4																								
2.																											
a)	Define encryption and decryption . Determine why roles are needed in case of privileges. Explain with appropriate assumption.	Aut-24	4																								
b)	Consider the following schemas for relations and answer the following questions: Customer (customerID, cname, email, phone, address, orderID) Order (orderID, productID, item, quantity, price, orderDate) Product (productID, name, model) i. Write the SQL statement to create these relations — you have to include all integrity constraints . ii. Define integrity constraints on Order relation where quantity must be less than 10 . iii. Determine the trigger statement for each order after the order placement of a product more than 5 quantity, where you have to give 5% discount on that product. Keep the log track for this trigger to see the discount price of that order. OR Normalize the following table if it is not normalized: <table><tr><th>pid</th><th>pname</th><th>emp_id</th><th>emp_name</th><th>dept_id</th><th>dept_name</th></tr><tr><td>P01</td><td>Alpha</td><td>E01</td><td>John</td><td>D01</td><td>HR</td></tr><tr><td>P02</td><td>Beta</td><td>E02</td><td>Alice</td><td>D02</td><td>Finance</td></tr><tr><td>P01</td><td>Alpha</td><td>E03</td><td>Mike</td><td>D01</td><td>HR</td></tr></table>	pid	pname	emp_id	emp_name	dept_id	dept_name	P01	Alpha	E01	John	D01	HR	P02	Beta	E02	Alice	D02	Finance	P01	Alpha	E03	Mike	D01	HR	Aut-24	6
pid	pname	emp_id	emp_name	dept_id	dept_name																						
P01	Alpha	E01	John	D01	HR																						
P02	Beta	E02	Alice	D02	Finance																						
P01	Alpha	E03	Mike	D01	HR																						

3.																																			
a)	Consider the following relational database: Patient(pid, pname, Address, mobile, DOB, gender) visit(pid, did, visit_date) Doctor(did, dname, speciality) drug(drid, d_name, d_type, manuf_year, unit_price) prescribe(pid, did, drid, pdate, quantity) Give an SQL DDL definition of this database. Identify referential-integrity constraints that should hold, and include them in the DDL definition. Ensure the following constraints: • pname, dname, and d_name → NOT NULL • mobile → UNIQUE • gender → only ‘M’ or ‘F’ • quantity and unit_price → not negative	Sp-24	6																																
a)	OR Normalize the following table if it is not normalized: <table><tr><th>sid</th><th>sname</th><th>address</th><th>Course_id</th><th>cname</th><th>Department_Id</th><th>dname</th><th>D_location</th></tr><tr><td>S01</td><td>Forhad</td><td>ctg</td><td>C01, C02</td><td>C, C++</td><td>D01</td><td>CSE</td><td>GEC</td></tr><tr><td>S02</td><td>Anika</td><td>dhaka</td><td>C01, C03</td><td>C, JAVA</td><td>D01</td><td>CSE</td><td>GEC</td></tr><tr><td>S03</td><td>Mazed</td><td>ctg</td><td>B01</td><td>Statistics</td><td>D02</td><td>BBA</td><td>Agrabad</td></tr></table>	sid	sname	address	Course_id	cname	Department_Id	dname	D_location	S01	Forhad	ctg	C01, C02	C, C++	D01	CSE	GEC	S02	Anika	dhaka	C01, C03	C, JAVA	D01	CSE	GEC	S03	Mazed	ctg	B01	Statistics	D02	BBA	Agrabad	Sp-24	6
sid	sname	address	Course_id	cname	Department_Id	dname	D_location																												
S01	Forhad	ctg	C01, C02	C, C++	D01	CSE	GEC																												
S02	Anika	dhaka	C01, C03	C, JAVA	D01	CSE	GEC																												
S03	Mazed	ctg	B01	Statistics	D02	BBA	Agrabad																												
b)	How does a trigger ensure integrity of the database? Give example. OR Write an assertion for the patient database to ensure that the patient is not visiting two doctors at a time. (May consider the schema of Q1(a)).	Sp-24	4																																
4.																																			
a)	Describe encryption , decryption , and privilege .	Aut-23	2																																
b)	Given the following schema: Client (cid, cname) Contract (cn_id, client_id, contract_title, dateFrom, dateTo) Write an assertion that will allow only one contract for a client from “19-Jun-2020” to “25-Jun-2020”.	Aut-23	3																																

c)	Given the following tables of information: Employee (ID, cname, Age, Address, Salary, designation, dept) Salary_pay (pid, ID, salary_of_month, pay_date) Write a trigger that will automatically increase the salary (5%) of the employee if he is from the "Finance" department and receiving the salary of June 2020 and onwards. OR What are the different types of referential integrity constraints, and how can they be used to ensure the integrity of data in a relational database?	Aut-23	5
5.			
a)	Consider the following database schema and answer the questions: Employee (E_id, C_id, E_Name, Email, Phone_no, Salary, Address, F_id) Customer (C_ID, C_Name, Email, Phone_no, F_id) Food_Item (F_ID, C_ID, F_Name, Price, Quantity) 1. Write the SQL statements required to create these relations. You have to include all integrity constraints. 2. Define constraints on the Food_Item table such that quantity should be at least one item. 3. Define constraints on Employee such that a minimum 15000 BDT pay scale is followed by the organization.	Sp-23	6
b)	i) Define referential integrity constraints and domain constraints with suitable examples. ii) Describe the following: a) Encryption and Decryption b) Privileges	Sp-23	4
6.			
a)	Consider the following relational schema and briefly answer the questions that follow: Emp (eid: integer, ename: string, age: integer, salary: real) Works (eid: integer, did: integer, pct_time: integer) Dept (did: integer, budget: real, managerid: integer) 1. Write the SQL statements required to create these relations, including all integrity constraints such as primary keys and foreign keys. 2. Define a domain constraint on Emp that will ensure that every employee makes at least \$10,000. 3. Define an assertion on Dept that ensures that all managers have age > 30.	Aut-22	6

b)	Define referential integrity. Explain the tests that must be made to preserve referential integrity for delete operation.	Aut-22	4
-----------	---	---------------	----------